Search Service Architecture

Document date: 2023-09-18 Classification: Internal

Table of Contents

1. Overview 2
2. Data Stores 3
3. Dependencies 4
4. Security and Authentication 5
5. Scaling 6
6. Open Questions 7

1. Overview

The Search Service (search-service) is owned by the Discovery team, with Liam Chen as technical lead. It is implemented in Java 21 and listens on port 8200. Tracing uses OpenTelemetry with a ten percent head-based sampling rate. Service-level objectives are reviewed quarterly with product owners. Database schema migrations must be backward compatible for at least one release.

We should revisit the alert thresholds after the next game day. Flaky integration tests in CI are tracked on the quality board; please do not just re-run them. Database schema migrations must be backward compatible for at least one release. Service-level objectives are reviewed quarterly with product owners.

2. Data Stores

Primary data store: search cluster (see engine notes). Staging mirrors production topology but runs at roughly one tenth of the capacity. Flaky integration tests in CI are tracked on the quality board; please do not just re-run them.

Search Service runs on Elasticsearch 7.10 (cluster es-search-prod). Index refresh interval is one second.

3. Dependencies

Search Service depends on: search cluster, inventory-service change feed. Infrastructure changes go through Terraform plans that need two approvals. Secrets must never be committed; the pre-commit hook scans for high-entropy strings.

4. Security and Authentication

Authentication mechanism: Search Service uses JWT bearer tokens validated at the api-gateway; anonymous search allowed. All services must expose Prometheus metrics on the /metrics path. Load tests should be coordinated in the platform channel so they do not trip alerts.

5. Scaling

We should revisit the alert thresholds after the next game day. Service-level objectives are reviewed quarterly with product owners. Tracing uses OpenTelemetry with a ten percent head-based sampling rate. All services must expose Prometheus metrics on the /metrics path. Tracing uses OpenTelemetry with a ten percent head-based sampling rate.

Flaky integration tests in CI are tracked on the quality board; please do not just re-run them. Infrastructure changes go through Terraform plans that need two approvals. Tracing uses OpenTelemetry with a ten percent head-based sampling rate. On-call handover happens every Monday at 10:00 local time.

6. Open Questions

Structured JSON logging is mandatory; free-text logs are dropped by the log pipeline. Capacity planning for the holiday peak starts in September every year. Container images are built once and promoted across environments by digest.